

everyone. Infrastructure resides under the control of the public cloud service provider, and the customers don't have any control over data storage location, customisation, etc. Some of the examples of public cloud services are *E-mail*, *Google Docs*, etc.

Private cloud: This cloud services setup is done as per the organisation's need. It provides customisation and secure access along with scalability as the business grows.

Hybrid cloud: A hybrid cloud is an amalgamation of public or private cloud services and on-premise infrastructure. This provides an option to utilise best-of-breed services among those available from various CSPs. It requires an effective level of integration between the services, and enforcement of the required security and management.

Community cloud: This is a shared infrastructure specific to a community. It is a multi-tenant shared common platform between

communities of similar needs, concerns, security, compliance, etc. Government departments, universities, and central banks are some of the examples of community cloud users.

Evaluation parameters of solutions

Figure 3 gives the parameters that need to be considered before opting for a cloud computing solution.

- **Computing:** The enterprise's computing needs like managed instances, edge computing, serverless computing, etc.
- **Storage and database:** The enterprise's storage and database needs like structured and unstructured data, archival storage, data lake, etc.
- **Infrastructure and networking:** The organisation's infrastructure and networking needs like virtual network, gateways, VPN, etc.
- **Application services:** The enterprise's application services

needs like analytics, development tools, integration tools, etc.

- **Management and governance:** The enterprise's management and governance needs like application monitoring and logs, infrastructure monitoring, usage reports, alter/notification, configurations, etc.
- **Security, identity and compliance:** The enterprise's security, identity and compliance needs like authentication, authorisation, role and privilege management, compliance enablement, security enforcement, etc.

These defined parameters consider various functional and non-functional aspects as well.

- **Non-functional aspects –** Availability, recoverability, resilience, security, data integrity, capacity, manageability, scalability, threat management, usability, performance, reliability, maintainability, interoperability, risk management, survivability, disaster recovery, regulatory, etc.
- **Functional aspects –** Application features, transactions, administration, authentication, authorisation, tracking, external interfaces, compliance, reporting, certifications, historical data, governance, business rules, audit, legal or regulatory, etc.

Hybrid cloud and multi-cloud

No one size fits all cloud solutions, and single cloud providers can't be everything for everyone. Multi-cloud and hybrid cloud are the two different architectural approaches for cloud solutions. In the multi-cloud model, infrastructure/services reside on more than one public cloud provider, while the hybrid cloud is an orchestration between on-premise (private cloud) systems and public cloud services.

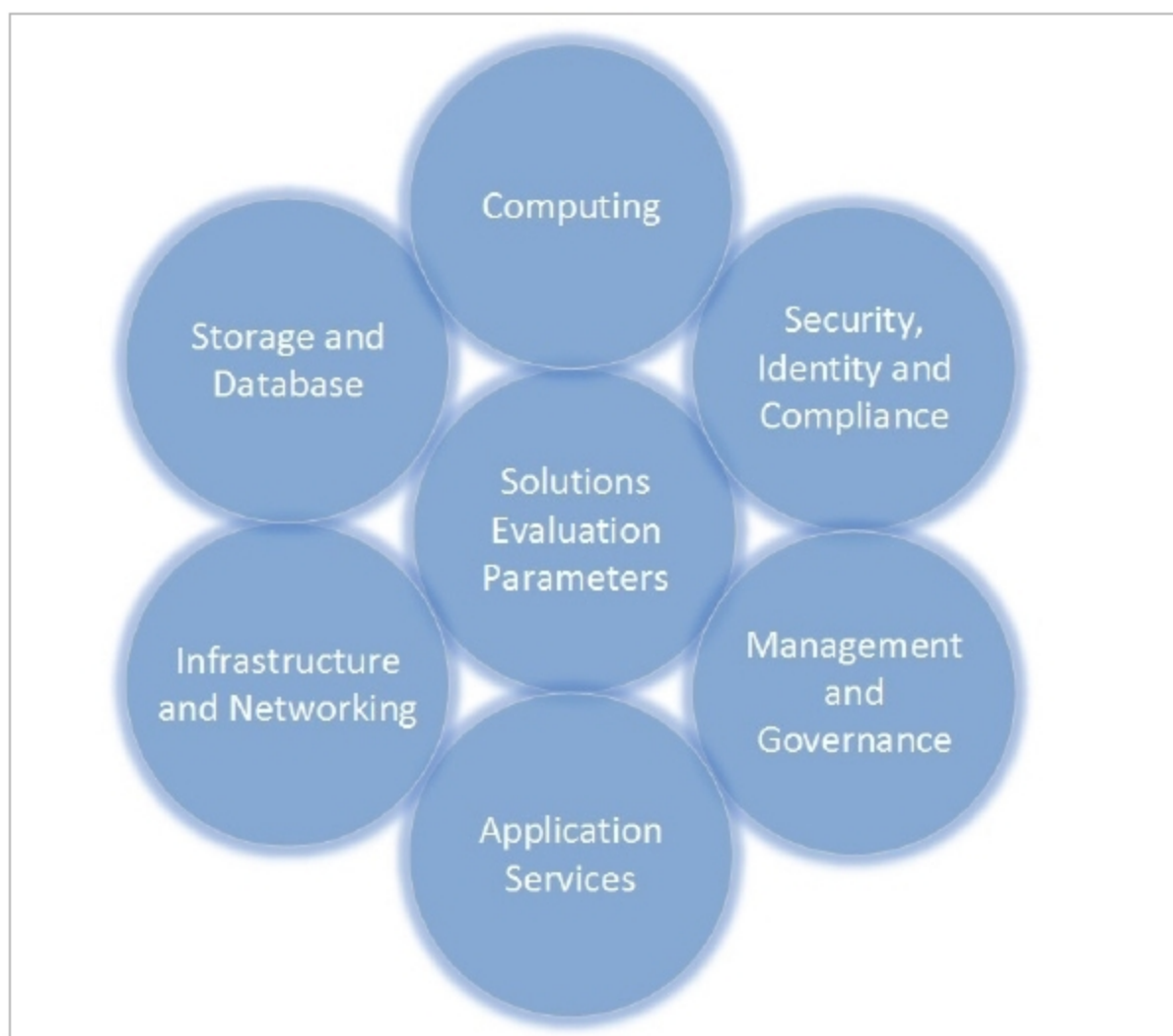


Figure 3: Parameters for consideration before choosing a cloud computing solution